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Touch panel with connection channel of adjacent touch electrode blocks and display device

Abstract

A touch panel includes: a first touch electrode including a first corner protrusion and first side protrusions including an adjacent-type first side protrusion, and a second touch electrode including a second corner protrusion and second side protrusions including an adjacent-type second side protrusion. The first side protrusion includes a second side sub-protrusion. The first corner protrusion includes a second corner sub-protrusion. The second side protrusion includes a fourth side sub-protrusion. The second corner protrusion includes a fourth corner sub-protrusion. The area of at least one of the second corner sub-protrusion and the second side sub-protrusion of the adjacent-type first side protrusion is smaller than that of the other second side sub-protrusion; and/or the area of at least one of the fourth corner sub-protrusion and the fourth side sub-protrusion of the adjacent-type second side protrusion is smaller than that of the other fourth side sub-protrusion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/033,607, which is a National Stage of International Application No. PCT/CN2021/097264, filed May 31, 2021, the entire content of which is hereby incorporated by reference.

FIELD

(1) The invention relates to the field of semiconductor technology, and in particular to a touch panel and a display device.

BACKGROUND

(2) In a touch panel, detection of a touch position is realized by arranging the transmitting (Tx) electrodes and receiving (Rx) electrodes crosswise on the panel along mutually orthogonal directions.

BRIEF SUMMARY

(3) Embodiments of the disclosure provide a touch panel and a display device. The touch panel includes a first touch electrode and a second touch electrode.

(4) The first touch electrode includes a plurality of first touch electrode blocks sequentially arranged in a first direction and electrically connected. The first touch electrode block includes a first main body, first side protrusions located on a side of the first main body and extending out in a direction perpendicular to the side, and a first corner protrusion located in a corner region of the first main body and extending out in a direction perpendicular to a side where the first corner protrusion is located. Two adjacent first touch electrode blocks arranged in the first direction are electrically connected through first corner protrusions. The first side protrusions include: an adjacent-type first side protrusion adjacent to the first corner protrusion and located on a side different from the side where the first corner protrusion is located. The first side protrusion includes: a first side sub-protrusion and a second side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first side protrusion is located. The first corner protrusion includes: a first corner sub-protrusion and a second corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first corner protrusion is located.

(5) The second touch electrode includes a plurality of second touch electrode blocks sequentially arranged in a second direction and electrically connected. The second touch electrode is cross-insulated from the first touch electrode. The second touch electrode block includes a second main body, second side protrusions located on a side of the second main body and extending out in a direction perpendicular to the side, and a second corner protrusion located in a corner region of the second main body and extending out in a direction perpendicular to a side where the second corner protrusion is located. Two adjacent second touch electrode blocks arranged in the second direction are electrically connected through second corner protrusions. The second side protrusions include: an adjacent-type second side protrusion adjacent to the second corner protrusion and located on a side different from the side where the second corner protrusion is located. The second side protrusion includes: a third side sub-protrusion and a fourth side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second side protrusion is located. The second corner protrusion includes: a third corner sub-protrusion and a fourth corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second corner protrusion is located. An area of at least one of the second corner sub-protrusion and the second side sub-protrusion of the adjacent-type first side protrusion is smaller than an area of the second side sub-protrusion of other first side protrusion; and/or, an area of at least one of the fourth corner sub-protrusion and the fourth side sub-protrusion of the adjacent-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion.

(6) In possible embodiments, a height of the second corner sub-protrusion in the direction perpendicular to the side where the first corner protrusion is located is smaller than a height of the second side sub-protrusion in the direction perpendicular to the side where the first side protrusion is located, so that an area of the second corner sub-protrusion is smaller than the area of the second side sub-protrusion of other first side protrusion; and/or, a height of the fourth corner sub-protrusion in the direction perpendicular to the side where the second corner protrusion is located is smaller than a height of the fourth side sub-protrusion in the direction perpendicular to the side where the second side protrusion is located, so that an area of the fourth corner sub-protrusion is smaller than the area of the fourth side sub-protrusion of other second side protrusion.

(7) In possible embodiments, the height of the second corner sub-protrusion in the direction perpendicular to the side where the first corner protrusion is located is one-sixth to one-half of the

height of the second side sub-protrusion in the direction perpendicular to the side where the first side protrusion is located; and the height of the fourth corner sub-protrusion in the direction perpendicular to the side where the second corner protrusion is located is one-sixth to one-half of the height of the fourth side sub-protrusion in the direction perpendicular to the side where the second side protrusion is located.

(8) In possible embodiments, a side of the second corner sub-protrusion facing an adjacent first touch electrode block has a first excision portion; and/or, a side of the fourth corner sub-protrusion facing an adjacent second touch electrode block has a second excision portion.

(9) In possible embodiments, a side of the second side sub-protrusion of the adjacent-type first side protrusion facing an adjacent first touch electrode block has a third excision portion; and/or, a side of the fourth side sub-protrusion of the adjacent-type second side protrusion facing an adjacent second touch electrode block has a fourth excision portion.

(10) In possible embodiments, a width of the second side sub-protrusion of the adjacent-type first side sub-protrusion in a direction parallel to the side where the adjacent-type first side sub-protrusion is located is smaller than a width of other second side sub-protrusion in a direction parallel to the side where the other first side protrusion is located; and/or, a width of the fourth side sub-protrusion of the adjacent-type second side sub-protrusion in a direction parallel to the side where the adjacent-type second side sub-protrusion is located is smaller than a width of other fourth side sub-protrusion in a direction parallel to the side where the other second side protrusion is located.

(11) In possible embodiments, a shape of the first excision portion is a triangle or a rectangle; a shape of the second excision portion is a triangle or a rectangle; a shape of the third excision portion is a triangle or a rectangle; and a shape of the fourth excision portion is a triangle or a rectangle.

(12) In possible embodiments, the first side protrusions include: a separated-type first side protrusion located on a same side as the adjacent-type first side protrusion and located at a corner adjacent to a corner where the first corner protrusion is located; and an area of the second side sub-protrusion of the separated-type first side protrusion is smaller than an area of the second side sub-protrusion of other first side protrusion; and/or, the second side protrusions include: a separated-type second side protrusion located on a same side as the adjacent-type second side protrusion and located at a corner adjacent to a corner where the second corner protrusion is located; and an area of the fourth side sub-protrusion of the separated-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion.

(13) In possible embodiments, a height of the second side sub-protrusion of the separated-type first side protrusion in a direction perpendicular to the side where the separated-type first side protrusion is located is smaller than a height of the second side sub-protrusion of other first side protrusion in the direction perpendicular to the side where the other first side protrusion is located; and a height of the fourth side sub-protrusion of the separated-type second side protrusion in a direction perpendicular to the side where the separated-type second side protrusion is located is smaller than a height of the fourth side sub-protrusion of other second side protrusion in the direction perpendicular to the side where the other second side protrusion is located.

(14) In possible embodiments, the height of the second side sub-protrusion of the separated-type first side protrusion in the direction perpendicular to the side where the separated-type first side protrusion is located is one-sixth to one-half of the height of the second side sub-protrusion of other first side protrusion in the direction perpendicular to the side where the other first side protrusion is located; and the height of the fourth side sub-protrusion of the separated-type second side protrusion in the direction perpendicular to the side where the separated-type second side protrusion is located is one-sixth to one-half of the height of the fourth side sub-protrusion of other second side protrusion in the direction perpendicular to the side where the other second side protrusion is located.

- (15) In possible embodiments, a side of the separated-type first side protrusion facing an adjacent first touch electrode has a fifth excision portion; and/or, a side of the separated-type second side protrusion facing an adjacent second touch electrode has a sixth excision portion.
- (16) In possible embodiments, a shape of the fifth excision portion is a triangle or a rectangle; and a shape of the sixth excision portion is a triangle or a rectangle.
- (17) In possible embodiments, a width of the second side sub-protrusion in a direction parallel to the side where the first side protrusion is located is smaller than a width of the first side sub-protrusion in a direction parallel to the side where the first side protrusion is located; and a width of the second corner sub-protrusion in a direction parallel to the side where the first corner protrusion is located is smaller than a width of the first corner sub-protrusion in a direction parallel to the side where the first corner protrusion is located.
- (18) In possible embodiments, the first corner protrusion includes: a first corner extension portion extending from the first corner sub-protrusion to an adjacent first touch electrode block in the direction parallel to the side where the first corner protrusion is located, and two adjacent first touch electrode blocks are electrically connected through first corner extension portions; and the second corner protrusion includes: a second corner extension portion extending from the second corner sub-protrusion to an adjacent second touch electrode block in the direction parallel to the side where the second corner protrusion is located, and two adjacent second touch electrode blocks are electrically connected through second corner extension portions.
- (19) In possible embodiments, the first touch electrode block and the second touch electrode block include a plurality of metal wires that form a metal grid, and boundaries of the first touch electrode block and the second touch electrode block are segmented by broken lines of the metal grid.
- (20) In possible embodiments, at least part of the metal grid inside the first corner protrusion is disconnected; and at least part of the metal grid inside the second corner protrusion is disconnected.
- (21) In possible embodiments, the first touch electrode blocks of the same first touch electrode are integrally connected; and the second touch electrode blocks of the same second touch electrode are bridged by a bridging portion.
- (22) In possible embodiments, the first touch electrode and the second touch electrode are in a same layer and made of a same material, there is an insulating layer between the bridging portion and a layer where the second touch electrode blocks are located, and two adjacent second touch electrode blocks are connected by the bridging portion through a via hole penetrating the insulating layer.
- (23) In possible embodiments, material of the bridging portion is same as material of the second touch electrode.
- (24) In possible embodiments, a plurality of floating electrodes are provided inside the first main body and the second main body.
- (25) Embodiments of the disclosure further provide a display device, including a display panel, and the touch panel as provided in the embodiments of the disclosure located on a light-emitting side of the display panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic top view of a touch display panel according to an embodiment of the disclosure.
- (2) FIG. 2A is a first enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (3) FIG. 2B is a second enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (4) FIG. 3 is a third enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (5) FIG. 4A is a fourth enlarged schematic diagram of FIG. 1 at the dotted line box S.

- (6) FIG. 4B is a fifth enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (7) FIG. 5A is a sixth enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (8) FIG. 5B is a seventh enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (9) FIG. 6A is an eighth enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (10) FIG. 6B is a ninth enlarged schematic diagram of FIG. 1 at the dotted line box S.
- (11) FIG. 7 is a partially enlarged schematic diagram of a connection channel of the second touch electrode of FIG. 4A.
- (12) FIG. 8 is a schematic cross-sectional view of a touch panel according to an embodiment of the disclosure.
- (13) FIG. 9 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure.
- (14) FIG. 10 is a schematic diagram of a relationship between the first touch electrode and the anode according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (15) In order to make the purposes, technical solutions and advantages of the embodiments of the present disclosure clearer, the technical solutions of the embodiments of the present disclosure will be described clearly and completely below in combination with the accompanying drawings of the embodiments of the present disclosure. Obviously the described embodiments are a part of the embodiments of the disclosure but not all the embodiments. Based upon the described embodiments of the disclosure, all of other embodiments obtained by those ordinary skilled in the art without creative work pertain to the protection scope of the disclosure.
- (16) Unless otherwise defined, the technical or scientific terms used in the disclosure shall have the general meaning understood by those ordinary skilled in the art to which the disclosure belongs. The “first”, “second” and similar words used in the disclosure do not represent any order, quantity or importance, and are only used to distinguish different components. The word such as “include” or “contain” or the like means that the element or object appearing before this word encompasses the elements or objects and their equivalents listed after this word, without excluding other elements or objects. The word such as “connect” or “connected” or the like is not limited to the physical or mechanical connection, but can include the electrical connection, whether direct or indirect. The words such as “up”, “down”, “left”, “right” are only used to represent the relative position relationship. When the absolute position of a described object changes, the relative position relationship may also change accordingly.
- (17) In order to keep the following description of the embodiments of the disclosure clear and concise, the disclosure omits the detailed description of known functions and known components.
- (18) The electrode pattern in the display region (AA) is composed of electrode pattern units. In the electrode layout scheme proposed in the Flexible Multiple Layer on Cell (FMLOC) technology, the Tx electrodes and the Rx electrodes are arranged on the same layer, and the electrode pattern unit consists of three parts: Tx (transmitting electrode), Rx (receiving electrode) and Dummy (non-effect region). In order to increase the mutual capacitance value between the Tx and Rx electrodes and enhance the touch performance, boundaries of Tx and Rx are generally designed as a type of “custom character”. The characters of “custom character” are designed to be the same in size, and the characters of “custom character” for Tx and Rx are inlaid with each other, so as to increase the area of the junction of Tx and Rx. However, due to factors such as pixel size and pattern unit size, the connectivity of the electrode pattern in some regions is poor, which increases the risk of subsequent poor touch control.
- (19) In view of this, referring to FIG. 1, FIG. 2A, FIG. 2B and FIG. 4A, where FIG. 2A, FIG. 2B and FIG. 4A are enlarged schematic diagrams of the wireframe S in FIG. 1, embodiments of the disclosure provide a touch panel including first touch electrodes 1 and second touch electrodes 2.
- (20) The first touch electrode 1 includes a plurality of first touch electrode blocks 10 sequentially arranged in a first direction A1 and electrically connected. As shown in FIG. 2A, the first touch

electrode block **10** includes a first main body **100**, first side protrusions **11** located on each side of the first main body **100** and extending out in a direction (as shown by the arrow **B1** in FIG. 2A) perpendicular to the each side, and a first corner protrusion **12** located in a corner region of the first main body **100** and extending out in a direction (as shown by the arrow **B1** in FIG. 2A) perpendicular to a side of the first main body **100** where the first corner protrusion **12** is located. Two adjacent first touch electrode blocks **10** arranged in the first direction **A1** are electrically connected through first corner protrusions **12**. The first side protrusions **11** include an adjacent-type first side protrusion **1101**, adjacent to the first corner protrusion **12** and located on a side of the first main body **100** different from the side of the first main body **100** where the first corner protrusion **12** is located. The first side protrusion **11** includes: a first side sub-protrusion **111** and a second side sub-protrusion **112** sequentially stacked in the direction perpendicular to the side of the first main body **100** where the first side protrusion **11** is located and away from the side (as shown by the arrow **B1** in FIG. 2A). The first corner protrusion **12** includes: a first corner sub-protrusion **121** and a second corner sub-protrusion **122** sequentially stacked in the direction perpendicular to the side of the first main body **100** where the first corner protrusion **12** is located and away from the side (as shown by the arrow **B1** in FIG. 2A).

(21) The second touch electrode **2** includes a plurality of second touch electrode blocks **20** sequentially arranged in a second direction **A2** and electrically connected. The second touch electrodes **2** and the first touch electrodes **1** are crosswise arranged and insulated. As shown in FIG. 2B, the second touch electrode block **20** includes a second main body **200**, second side protrusions **21** located on each side of the second main body **200** and extending out in a direction (as shown by the arrow **B2** in FIG. 2B) perpendicular to the each side, and a second corner protrusion **22** located in a corner region of the second main body **200** and extending out in a direction (as shown by the arrow **B2** in FIG. 2B) perpendicular to a side of the second main body **200** where the second corner protrusion **22** is located. Two adjacent second touch electrode blocks **20** arranged in the second direction **A2** are electrically connected through second corner protrusions **22**. The second side protrusions **21** include an adjacent-type second side protrusion **2101** adjacent to the second corner protrusion **22** and located on a side of the second main body **200** different from the side of the second main body **200** where the second corner protrusion **22** is located. The second side protrusion **21** includes: a third side sub-protrusion **211** and a fourth side sub-protrusion **212** sequentially stacked in the direction perpendicular to the side of the second main body **200** where the second side protrusion **21** is located and away from the side (as shown by the arrow **B2** in FIG. 2B). The second corner protrusion **22** includes: a third corner sub-protrusion **221** and a fourth corner sub-protrusion **222** sequentially stacked in the direction perpendicular to the side of the second main body **200** where the second corner protrusion **22** is located and away from the side (as shown by the arrow **B2** in FIG. 2B). An area of at least one of the second corner sub-protrusion **222** and the second side sub-protrusion **212** of the adjacent-type first side protrusion **1101** is smaller than an area of the second side sub-protrusion **112** of other first side protrusion **11**; and/or, an area of at least one of the fourth corner sub-protrusion **222** and the fourth side sub-protrusion **212** of the adjacent-type second side protrusion **2101** is smaller than an area of the fourth side sub-protrusion **212** of other second side protrusion **21**.

(22) In the embodiments of the disclosure, the first touch electrode block **10** includes first side protrusions **11** and a first corner protrusion **12** connecting two adjacent first touch electrode blocks **10**. The first side protrusion **11** includes a second side sub-protrusion **112**. The first corner protrusion **12** includes a second corner sub-protrusion **122**. The second touch electrode **20** includes second side protrusions **21** and a second corner protrusion **22** connecting two adjacent second touch electrode blocks **20**. The area of at least one of the second corner sub-protrusion **122** and the second side sub-protrusion **112** of the adjacent-type first side protrusion **1101** is smaller than the area of the second side sub-protrusion **112** of other first side protrusion **11**, so that the width of the region between the second corner sub-protrusion **122** and the first side protrusion **11** of the opposite

first touch electrode block **10** (the region as shown by the dotted circle S3 in FIG. 2A) can be increased, that is, the width of the connection channel of the adjacent second touch electrode blocks **20** is increased, and the problems of poor connectivity and high resistance when the connection channel of the adjacent second touch electrode blocks **20** is relatively narrow are improved. The second side protrusion **21** includes a fourth side sub-protrusion **212**. The second corner protrusion **22** includes a fourth corner sub-protrusion **222**. And/or, the area of at least one of the fourth corner sub-protrusion **222** and the fourth side sub-protrusion **212** of the adjacent-type second side protrusion **2101** is smaller than the area of the fourth side sub-protrusion **212** of other second side protrusion **21**, so that the width of the region between the fourth corner sub-protrusion **222** and the second side protrusion **21** of the opposite second touch electrode block **20** (the region as shown by the dotted circle S4 in FIG. 2B) can be increased, that is, the width of the connection channel of the adjacent first touch electrode blocks **10** is increased, and the problems of poor connectivity and high resistance when the connection channel of the adjacent first touch electrode blocks **10** is relatively narrow are improved.

(23) In possible embodiments, as shown in FIG. 2A, the height h1 of the second corner sub-protrusion **122** in the direction perpendicular to the side is smaller than the height h2 of the second side sub-protrusion **112** in the direction perpendicular to the side, so that the area of the second corner sub-protrusion **122** is smaller than the area of the second side sub-protrusion **122** of other first side protrusion **11** (except the adjacent-type first side protrusion **1101**). That is, the distance between the end of the second corner sub-protrusion **122** and the dotted line S1 is greater than the distance between the end of the second side sub-protrusion **112** and the dotted line S1, so that the area of the second corner sub-protrusion **122** can be smaller than the area of the second side sub-protrusion **112**, and the pattern is easy to manufacture, simplifying the manufacturing process of the touch display panel. And/or, as shown in FIG. 2B, the height h3 of the fourth corner sub-protrusion **222** in the direction perpendicular to the side is smaller than the height h4 of the fourth side sub-protrusion **212** in the direction perpendicular to the side, so that the area of the fourth corner sub-protrusion **222** is smaller than the area of the fourth side sub-protrusion **222** of other second side protrusion **21** (except the adjacent-type second side protrusion **2101**). That is, the distance between the end of the fourth corner sub-protrusion **222** and the dotted line S2 is greater than the distance between the end of the fourth side sub-protrusion **212** and the dotted line S2, so that the area of the fourth corner sub-protrusion **222** is smaller than the area of the fourth side sub-protrusion **212**, and the pattern is easy to manufacture, simplifying the manufacturing process of the touch display panel.

(24) For example, the height h1 of the second corner sub-protrusion **122** in the direction perpendicular to the side is one-sixth to one-half of the height h2 of the second side sub-protrusion **112** in the direction perpendicular to the side, thus improving the problems of poor connectivity and high resistance of the adjacent second touch electrode blocks **20** while also making the area of the junction between the first touch electrode block **10** and the second touch electrode block **20** larger, so that there is a larger mutual capacitance value between the first touch electrode block **10** and the second touch electrode block **20**, and the touch panel has the enhanced touch performance. The height h3 of the fourth corner sub-protrusion **222** in the direction perpendicular to the side is one-sixth to one-half of the height h4 of the fourth side sub-protrusion **212** in the direction perpendicular to the side, thus improving the problems of poor connectivity and high resistance of the adjacent first touch electrode blocks **10** while also making the area of the junction between the first touch electrode block **10** and the second touch electrode block **20** larger, so that there is a larger mutual capacitance value between the first touch electrode block **10** and the second touch electrode block **20**, and the touch panel has the enhanced touch performance.

(25) In possible embodiments, as shown in FIG. 3, a side of the second corner sub-protrusion **122** facing an adjacent first touch electrode block **10** has a first excision portion K1. The region where the first excision portion K1 is located is used to arrange the second touch electrode block **20**, so

that the area of the second corner sub-protrusion **122** can be smaller than the area of the second side sub-protrusion **112**, and the pattern is easy to manufacture, simplifying the manufacturing process of the touch display panel. And/or, a side of the fourth corner sub-protrusion **222** facing an adjacent second touch electrode block **20** has a second excision portion **K2**. The region where the second excision portion **K2** is located is used to arrange the first touch electrode block **10**, so that the area of the fourth corner sub-protrusion **222** is smaller than the area of the fourth side sub-protrusion **212**, and the pattern is easy to manufacture, simplifying the manufacturing process of the touch display panel.

(26) In possible embodiments, as shown in FIG. 4A, a side of the second side sub-protrusion **112** of the adjacent-type first side protrusion **1101** facing an adjacent first touch electrode block **10** has a third excision portion **K3**, thus further increasing the width of the connection channel of the adjacent second touch electrode blocks **20**. And/or, a side of the fourth side sub-protrusion **212** of the adjacent-type second side protrusion **2101** facing an adjacent second touch electrode block **20** has a fourth excision portion **K4**, thus further increasing the width of the connection channel of the adjacent first touch electrode blocks **10**.

(27) It should be noted that, in the embodiments of the disclosure, in order to clearly describe the pattern shapes of the adjusted second corner sub-protrusion **122** and fourth corner sub-protrusion **222**, the description is given in the case where the second corner sub-protrusion **122** has a first excision portion **K1** and the fourth corner sub-protrusion **222** has a second excision portion **K2**. In implementations, during the manufacturing process, the first excision portion **K1** is a part removed from the second corner sub-protrusion **122** and does not belong to the structure of the second corner sub-protrusion **122**, and the region where the first excision portion **K1** is located is used to manufacture and arrange the second touch electrode block **20**. Similarly, during the manufacturing process, the second excision portion **K2** is a part removed from the fourth corner sub-protrusion **222** and does not belong to the structure of the fourth corner sub-protrusion **222**, and the region where the second excision portion **K2** is located is used to arrange the first touch electrode block **10**. Similarly, during the manufacturing process, the third excision portion **K3** is a part removed from the adjacent-type first side protrusion **1101** and does not belong to the structure of the adjacent-type first side protrusion **1101**, and the region where the third excision portion **K3** is located is used to arrange the second touch electrode block **20**. Similarly, during the manufacturing process, the fourth excision portion **K4** is a part removed from the adjacent-type second side protrusion **2101** and does not belong to the structure of the adjacent-type second side protrusion **2101**, and the region where the fourth excision portion **K4** is located is used to arrange the first touch electrode block **10**.

(28) In possible embodiments, as shown in FIG. 4B, the width **a1** of the second side sub-protrusion **112** of the adjacent-type first side sub-protrusion **1101** in the direction parallel to the side is smaller than the width **a2** of other second side sub-protrusion **112** in the direction parallel to the side, thus further increasing the width of the connection channel of the adjacent second touch electrode blocks **20**. And/or, the width **a3** of the fourth side sub-protrusion **212** of the adjacent-type second side sub-protrusion **2101** in the direction parallel to the side is smaller than the width **a4** of other fourth side sub-protrusion **212** in the direction parallel to the side, thus further increasing the width of the connection channel of the adjacent first touch electrode blocks **10**.

(29) In possible embodiments, the shape of the first excision portion **K1** is a triangle, as shown in FIG. 3; or the shape of the first excision portion **K1** may also be a rectangle. The shape of the second excision portion **K2** is a triangle, as shown in FIG. 3; or the shape of the second excision portion **K2** may also be a rectangle. The shape of the third excision portion **K3** is a triangle, as shown in FIG. 4A; or the shape of the third excision portion **K3** may also be a rectangle. The shape of the fourth excision portion **K4** is a triangle, as shown in FIG. 4A; or the shape of the fourth excision portion **K4** may also be a rectangle.

(30) In possible embodiments, as shown in FIG. 5A, the first side protrusion **11** includes: a

separated-type first side protrusion **1102** located on a same side as the adjacent-type first side protrusion **1101** and located at a corner adjacent to a corner where the first corner protrusion **12** is located. The area of the second side sub-protrusion **112** of the separated-type first side protrusion **1102** is smaller than the area of the second side sub-protrusion **112** of other first side protrusion **11**. And/or, as shown in FIG. 5B, the second side protrusion **21** includes: a separated-type second side protrusion **2102** located on a same side as the adjacent-type second side protrusion **2101** and located at a corner adjacent to a corner where the second corner protrusion **22** is located. The area of the fourth side sub-protrusion **212** of the separated-type second side protrusion **2102** is smaller than the area of the fourth side sub-protrusion **212** of other second side protrusion **21**. In embodiments of the disclosure, the first side protrusion **11** includes the separated-type first side protrusion **1102**. The area of the second side sub-protrusions **112** of the separated-type first side protrusion **1102** is smaller than the area of the second side sub-protrusion **112** of other first side protrusion **11**, which can improve the problem of poor connectivity of the second touch electrode block **20** in the non-connected region of the corner, and improve the problem that a part of electrodes on the corner of the second touch electrode block **20** cannot be connected to the second touch electrode **2** and form a touch invalid region when the separated-type first side protrusion **1102** is the same in height as other first side protrusion **11** due to the “custom character” character design. The second side protrusion **21** includes the separated-type second side protrusion **2102**. The area of the fourth side sub-protrusions **212** of the separated-type second side protrusion **2102** is smaller than the area of the fourth side sub-protrusion **212** of other second side protrusion **21**, which can improve the problem of poor connectivity of the first touch electrode block **10** in the non-connected region of the corner, and improve the problem that a part of electrodes on the corner of the first touch electrode block **10** cannot be connected to the first touch electrode **1** and form a touch invalid region when the separated-type second side protrusion **2102** is the same in height as other second side protrusion **21** due to the “custom character” character design.

(31) In possible embodiments, as shown in FIG. 5A, the height $h5$ of the second side sub-protrusion **112** of the separated-type first side protrusions **1102** in the direction perpendicular to the side is smaller than the height $h6$ of the second side sub-protrusion **112** of other first side protrusion **11** in the direction perpendicular to the side, so that the area of the second side sub-protrusion **112** of the separated-type first side protrusion **1102** can be smaller than the area of the second side sub-protrusion **112** of other first side protrusion **11**. As shown in FIG. 5B, the height $h7$ of the fourth side sub-protrusion **212** of the separated-type second side protrusion **2102** in the direction perpendicular to the side is smaller than the height $h8$ of the fourth side sub-protrusion **212** of other second side protrusion **21** in the direction perpendicular to the side, so that the area of the fourth side sub-protrusion **212** of the separated-type second side protrusion **2102** can be smaller than the area of the fourth side sub-protrusion **212** of other second side protrusion **21**.

(32) In possible embodiments, as shown in FIG. 5A, the height $h5$ of the second side sub-protrusion **112** of the separated-type first side protrusions **1102** in the direction perpendicular to the side is one-sixth to one-half of the height $h6$ of the second side sub-protrusion **112** of other first side protrusion **11** in the direction perpendicular to the side, thus improving the problem of poor connectivity of the second touch electrode block **20** in the non-connected region of the corner while also making the area of the junction between the first touch electrode block **10** and the second touch electrode block **20** larger, so that there is a larger mutual capacitance value between the first touch electrode block **10** and the second touch electrode block **20**, and the touch panel has the enhanced touch performance. As shown in FIG. 5B, the height $h7$ of the fourth side sub-protrusion **212** of the separated-type second side protrusion **2102** in the direction perpendicular to the side is one-sixth to one-half of the height $h8$ of the fourth side sub-protrusion **212** of other second side protrusion **21** in the direction perpendicular to the side, thus improving the problem of poor connectivity of the first touch electrode block **10** in the non-connected region of the corner while also making the area of the junction between the first touch electrode block **10** and the second

touch electrode block **20** larger, so that there is a larger mutual capacitance value between the first touch electrode block **10** and the second touch electrode block **20**, and the touch panel has the enhanced touch performance.

(33) In possible embodiments, as shown in FIG. **6A**, a side of the separated-type first side protrusion **1102** facing an adjacent first touch electrode **1** has a fifth excision portion **K5**, thus improving the problem of poor connectivity of the second touch electrode block **20** in the non-connected region of the corner. And/or, as shown in FIG. **6B**, a side of the separated-type second side protrusion **2102** facing an adjacent second touch electrode **2** has a sixth excision portion **K6**, thus improving the problem of poor connectivity of the first touch electrode block **10** in the non-connected region of the corner.

(34) In possible embodiments, as shown in FIG. **6A**, the shape of the fifth excision portion **K5** is a triangle or a rectangle. As shown in FIG. **6B**, the shape of the sixth excision portion **K6** is a triangle or a rectangle.

(35) In possible embodiments, as shown in FIG. **2A**, the width **h9** of the second side sub-protrusion **112** in the direction parallel to the side is smaller than the width **h10** of the first side sub-protrusion **111** in the direction parallel to the side. As shown in FIG. **2A**, the width **h11** of the second corner sub-protrusion **122** in the direction parallel to the side is smaller than the width **h12** of the first corner sub-protrusion **121** in the direction parallel to the side.

(36) In possible embodiments, as shown in FIG. **2A**, the first corner protrusion **12** includes: a first corner extension portion **123** extending from the first corner sub-protrusion **121** to an adjacent first touch electrode block **10** in the direction parallel to the side, and two adjacent first touch electrode blocks **10** are electrically connected through first corner extension portions **123**. As shown in FIG. **2B**, the second corner protrusion **22** includes: a second corner extension portion **223** extending from the second corner sub-protrusion **221** to an adjacent second touch electrode block **20** in the direction parallel to the side, and two adjacent second touch electrode blocks **20** are electrically connected through second corner extension portions **223**.

(37) In possible embodiments, as shown in FIG. **7**, the first touch electrode block **10** and the second touch electrode block **20** include a plurality of metal wires that form a metal grid. Boundaries of the first touch electrode block **10** and the second touch electrode block **20** are segmented by broken lines of the metal grid.

(38) In possible embodiments, as shown in FIG. **7**, at least part of the metal grid inside the first corner protrusion **12** is disconnected; and at least part of the metal grid inside the second corner protrusion **22** is disconnected.

(39) In possible embodiments, as shown in FIG. **1**, the first touch electrode blocks **10** of the same first touch electrode **1** are integrally connected; and the second touch electrode blocks **20** of the same second touch electrode **2** are bridged by a bridging portion **3**.

(40) In possible embodiments, as shown in FIG. **8** which is a schematic cross-sectional view of FIG. **1** along the dotted line **CC1**, the first touch electrode **1** and the second touch electrode **2** are in the same layer and made of the same material. There is an insulating layer **5** between the bridging portion **3** and a layer where the second touch electrode blocks **20** are located. Two adjacent second touch electrode blocks **20** are connected by the bridging portion **3** through a via hole **K** penetrating the insulating layer **5**.

(41) In possible embodiments, the material of the bridging portion **3** is the same as the material of the second touch electrode **2**.

(42) In possible embodiments, as shown in FIG. **2A**, a plurality of floating electrodes **4** are provided inside the first main body **10** and the second main body **20**. For example, the floating electrode **4** can be understood as a structure in which an electrode pattern is provided but no electrical signal is loaded to the electrode pattern during use.

(43) Embodiments of the disclosure further provide a display device, including a display panel, and the touch panel as provided in the embodiments of the disclosure located on a light-emitting side of

the display panel.

(44) For example, as shown in FIG. 9, the display device may include a base substrate **000**, a display layer **6** located between the base substrate **000** and the bridging portion **3**, and an encapsulation layer **7** located between the display layer **6** and the bridging portion **3**.

(45) For example, as shown in FIG. 10, the display layer **6** may include a light emitting device. The light emitting device may include an anode layer having a plurality of anodes **61**, and a pixel definition layer having pixel openings **62**. The metal grids constituting the first touch electrode **1** and the second touch electrode **2** correspond to the pixel openings **62** one by one.

(46) For example, the display layer **6** may include a driving layer for driving the light emitting device. The driving layer may include a pixel circuit for driving the light emitting device to emit light. The pixel circuit may include a thin film transistor and a capacitor. For example, the structure of the pixel circuit and the structure of the light emitting device may be the same as those of the organic light-emitting display panel and the quantum dot light-emitting display panel in the prior art, and will not be repeated in the disclosure.

(47) During implementations, in embodiments of the disclosure, the display device may be a mobile phone, a tablet, a television, a display, a laptop, a digital photo frame, a navigator, a watch, a bracelet or any other product or component with display function. All of other indispensable components of the display device should be understood by those ordinary skilled in the art to be included, and will be omitted here and should not be considered as limitations on the disclosure.

(48) Although the preferred embodiments of the invention have been described, those skilled in the art can make additional alterations and modifications to these embodiments once they learn about the basic creative concepts. Thus the attached claims are intended to be interpreted to include the preferred embodiments as well as all the alterations and modifications falling within the scope of the invention.

(49) Evidently those skilled in the art can make various modifications and variations to the embodiments of the invention without departing from the spirit and scope of the embodiments of the invention. Thus the invention is also intended to encompass these modifications and variations therein as long as these modifications and variations to the embodiments of the invention come into the scope of the claims of the invention and their equivalents.

Claims

1. A touch panel, comprising: a first touch electrode comprising: a plurality of first touch electrode blocks sequentially arranged in a first direction and electrically connected; wherein the first touch electrode block comprises: a first main body, wherein an outline of the first main body is substantially a quadrilateral, sides of the quadrilateral are sides of the first main body, and top corners of the quadrilateral are corner regions of the first main body, first side protrusions located on a side of the first main body and extending out in a direction perpendicular to the side, and a first corner protrusion located in a corner region of the first main body and extending out in a direction perpendicular to a side where the first corner protrusion is located; wherein two adjacent first touch electrode blocks arranged in the first direction are electrically connected through the first corner protrusions; the first side protrusions comprise: an adjacent-type first side protrusion, adjacent to the first corner protrusion and located on a side different from the side where the first corner protrusion is located; the first side protrusion comprises: a first side sub-protrusion and a second side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first side protrusion is located; and the first corner protrusion comprises: a first corner sub-protrusion and a second corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first corner protrusion is located; and a second touch electrode comprising: a plurality of second touch electrode blocks sequentially arranged in a second direction and electrically connected, wherein the first direction intersects with the second

direction; wherein the second touch electrode and the first touch electrode are crosswise arranged and insulated; the second touch electrode block comprises: a second main body, wherein an outline of the second main body is substantially a quadrilateral, sides of the quadrilateral are sides of the second main body, and top corners of the quadrilateral are corner regions of the second main body, second side protrusions located on a side of the second main body and extending out in a direction perpendicular to the side, and a second corner protrusion located in a corner region of the second main body and extending out in a direction perpendicular to a side where the second corner protrusion is located; wherein two adjacent second touch electrode blocks arranged in the second direction are electrically connected through the second corner protrusions; the second side protrusions comprise: an adjacent-type second side protrusion, adjacent to the second corner protrusion and located on a side different from the side where the second corner protrusion is located; the second side protrusion comprises: a third side sub-protrusion and a fourth side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second side protrusion is located; and the second corner protrusion comprises: a third corner sub-protrusion and a fourth corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second corner protrusion is located; wherein an area of at least one of the second corner sub-protrusion and the second side sub-protrusion of the adjacent-type first side protrusion is smaller than an area of the second side sub-protrusion of other first side protrusion; and/or, an area of at least one of the fourth corner sub-protrusion and the fourth side sub-protrusion of the adjacent-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion.

2. The touch panel according to claim 1, wherein: a height of the second corner sub-protrusion in the direction perpendicular to the side where the first corner protrusion is located is smaller than a height of the second side sub-protrusion in the direction perpendicular to the side where the first side protrusion is located, so that an area of the second corner sub-protrusion is smaller than the area of the second side sub-protrusion of other first side protrusion; and/or, a height of the fourth corner sub-protrusion in the direction perpendicular to the side where the second corner protrusion is located is smaller than a height of the fourth side sub-protrusion in the direction perpendicular to the side where the second side protrusion is located, so that an area of the fourth corner sub-protrusion is smaller than the area of the fourth side sub-protrusion of other second side protrusion.

3. The touch panel according to claim 1, wherein: a side of the second corner sub-protrusion facing an adjacent first touch electrode block has a first excision portion; and/or, a side of the fourth corner sub-protrusion facing an adjacent second touch electrode block has a second excision portion.

4. The touch panel according to claim 3, wherein: a side of the second side sub-protrusion of the adjacent-type first side protrusion facing an adjacent first touch electrode block has a third excision portion; and/or, a side of the fourth side sub-protrusion of the adjacent-type second side protrusion facing an adjacent second touch electrode block has a fourth excision portion.

5. The touch panel according to claim 3, wherein: a width of the second side sub-protrusion of the adjacent-type first side sub-protrusion in a direction parallel to the side where the adjacent-type first side sub-protrusion is located is smaller than a width of the second side sub-protrusion of other first side protrusion in a direction parallel to the side where the other first side protrusion is located; and/or, a width of the fourth side sub-protrusion of the adjacent-type second side sub-protrusion in a direction parallel to the side where the adjacent-type second side sub-protrusion is located is smaller than a width of the fourth side sub-protrusion of other second side protrusion in a direction parallel to the side where the other second side protrusion is located.

6. The touch panel according to claim 4, wherein: the first side protrusions comprise: a separated-type first side protrusion located on a same side as the adjacent-type first side protrusion and located at a corner adjacent to a corner where the first corner protrusion is located; wherein an area of the second side sub-protrusion of the separated-type first side protrusion is smaller than an area

of the second side sub-protrusion of other first side protrusion; and/or, the second side protrusions comprise: a separated-type second side protrusion located on a same side as the adjacent-type second side protrusion and located at a corner adjacent to a corner where the second corner protrusion is located; wherein an area of the fourth side sub-protrusion of the separated-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion.

7. The touch panel according to claim 6, wherein: a height of the second side sub-protrusion of the separated-type first side protrusion in a direction perpendicular to the side where the separated-type first side protrusion is located is smaller than a height of the second side sub-protrusion of other first side protrusion in the direction perpendicular to the side where the other first side protrusion is located; and a height of the fourth side sub-protrusion of the separated-type second side protrusion in a direction perpendicular to the side where the separated-type second side protrusion is located is smaller than a height of the fourth side sub-protrusion of other second side protrusion in the direction perpendicular to the side where the other second side protrusion is located.

8. The touch panel according to claim 6, wherein: a side of the separated-type first side protrusion facing an adjacent first touch electrode has a fifth excision portion; and/or, a side of the separated-type second side protrusion facing an adjacent second touch electrode has a sixth excision portion.

9. The touch panel according to claim 1, wherein: a width of the second side sub-protrusion in a direction parallel to the side where the first side protrusion is located is smaller than a width of the first side sub-protrusion in a direction parallel to the side where the first side protrusion is located; and a width of the second corner sub-protrusion in a direction parallel to the side where the first corner protrusion is located is smaller than a width of the first corner sub-protrusion in a direction parallel to the side where the first corner protrusion is located.

10. The touch panel according to claim 9, wherein: the first corner protrusion comprises: a first corner extension portion extending from the first corner sub-protrusion to an adjacent first touch electrode block in the direction parallel to the side where the first corner protrusion is located; wherein two adjacent first touch electrode blocks are electrically connected through the first corner extension portions; and the second corner protrusion comprises: a second corner extension portion extending from the third corner sub-protrusion to an adjacent second touch electrode block in the direction parallel to the side where the second corner protrusion is located; wherein two adjacent second touch electrode blocks are electrically connected through the second corner extension portions.

11. The touch panel according to claim 1, wherein the first touch electrode block and the second touch electrode block comprise a plurality of metal wires that form a metal grid, and boundaries of the first touch electrode block and the second touch electrode block are segmented by broken lines of the metal grid.

12. The touch panel according to claim 6, wherein at least part of the metal grid inside the first corner protrusion is disconnected; and at least part of the metal grid inside the second corner protrusion is disconnected.

13. The touch panel according to claim 1, wherein the first touch electrode blocks of the same first touch electrode are integrally connected; and the second touch electrode blocks of the same second touch electrode are bridged by a bridging portion.

14. The touch panel according to claim 13, wherein the first touch electrode and the second touch electrode are in a same layer and made of a same material, an insulating layer is located between the bridging portion and a layer where the second touch electrode blocks are located, and two adjacent second touch electrode blocks are connected by the bridging portion through a via hole penetrating the insulating layer.

15. The touch panel according to claim 1, wherein a plurality of floating electrodes are provided inside the first main body and the second main body.

16. A display device, comprising a display panel, and the touch panel according to claim 1 located

on a light emitting side of the display panel.

17. A touch panel, comprising: a first touch electrode comprising: a plurality of first touch electrode blocks sequentially arranged in a first direction and electrically connected; wherein the first touch electrode block comprises: a first main body, first side protrusions located on a side of the first main body and extending out in a direction perpendicular to the side, and a first corner protrusion located in a corner region of the first main body and extending out in a direction perpendicular to a side where the first corner protrusion is located; wherein two adjacent first touch electrode blocks arranged in the first direction are electrically connected through the first corner protrusions; the first side protrusions comprise: an adjacent-type first side protrusion, adjacent to the first corner protrusion and located on a side different from the side where the first corner protrusion is located; the first side protrusion comprises: a first side sub-protrusion and a second side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first side protrusion is located; and the first corner protrusion comprises: a first corner sub-protrusion and a second corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first corner protrusion is located; and a second touch electrode comprising: a plurality of second touch electrode blocks sequentially arranged in a second direction and electrically connected; wherein the second touch electrode and the first touch electrode are crosswise arranged and insulated; the second touch electrode block comprises: a second main body, second side protrusions located on a side of the second main body and extending out in a direction perpendicular to the side, and a second corner protrusion located in a corner region of the second main body and extending out in a direction perpendicular to a side where the second corner protrusion is located; wherein two adjacent second touch electrode blocks arranged in the second direction are electrically connected through the second corner protrusions; the second side protrusions comprise: an adjacent-type second side protrusion, adjacent to the second corner protrusion and located on a side different from the side where the second corner protrusion is located; the second side protrusion comprises: a third side sub-protrusion and a fourth side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second side protrusion is located; and the second corner protrusion comprises: a third corner sub-protrusion and a fourth corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second corner protrusion is located; wherein an area of at least one of the second corner sub-protrusion and the second side sub-protrusion of the adjacent-type first side protrusion is smaller than an area of the second side sub-protrusion of other first side protrusion; and/or, an area of at least one of the fourth corner sub-protrusion and the fourth side sub-protrusion of the adjacent-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion; wherein a width of the second side sub-protrusion of the adjacent-type first side sub-protrusion in a direction parallel to the side where the adjacent-type first side sub-protrusion is located is smaller than a width of the second side sub-protrusion of other first side protrusion in a direction parallel to the side where the other first side protrusion is located; and/or a width of the fourth side sub-protrusion of the adjacent-type second side sub-protrusion in a direction parallel to the side where the adjacent-type second side sub-protrusion is located is smaller than a width of the fourth side sub-protrusion of other second side protrusion in a direction parallel to the side where the other second side protrusion is located.

18. A touch panel, comprising: a first touch electrode comprising: a plurality of first touch electrode blocks sequentially arranged in a first direction and electrically connected; wherein the first touch electrode block comprises: a first main body, first side protrusions located on a side of the first main body and extending out in a direction perpendicular to the side, and a first corner protrusion located in a corner region of the first main body and extending out in a direction perpendicular to a side where the first corner protrusion is located; wherein two adjacent first touch electrode blocks arranged in the first direction are electrically connected through the first corner protrusions; the first side protrusions comprise: an adjacent-type first side protrusion, adjacent to the first corner

protrusion and located on a side different from the side where the first corner protrusion is located; and a separated-type first side protrusion located on a same side as the adjacent-type first side protrusion and located at a corner adjacent to a corner where the first corner protrusion is located; the first side protrusion comprises: a first side sub-protrusion and a second side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first side protrusion is located; and the first corner protrusion comprises: a first corner sub-protrusion and a second corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the first corner protrusion is located; and a second touch electrode comprising: a plurality of second touch electrode blocks sequentially arranged in a second direction and electrically connected; wherein the second touch electrode and the first touch electrode are crosswise arranged and insulated; the second touch electrode block comprises: a second main body, second side protrusions located on a side of the second main body and extending out in a direction perpendicular to the side, and a second corner protrusion located in a corner region of the second main body and extending out in a direction perpendicular to a side where the second corner protrusion is located; wherein two adjacent second touch electrode blocks arranged in the second direction are electrically connected through the second corner protrusions; the second side protrusions comprise: an adjacent-type second side protrusion, adjacent to the second corner protrusion and located on a side different from the side where the second corner protrusion is located; and a separated-type second side protrusion located on a same side as the adjacent-type second side protrusion and located at a corner adjacent to a corner where the second corner protrusion is located; the second side protrusion comprises: a third side sub-protrusion and a fourth side sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second side protrusion is located; and the second corner protrusion comprises: a third corner sub-protrusion and a fourth corner sub-protrusion sequentially stacked in the direction perpendicular to and away from the side where the second corner protrusion is located; wherein an area of at least one of the second corner sub-protrusion and the second side sub-protrusion of the adjacent-type first side protrusion is smaller than an area of the second side sub-protrusion of other first side protrusion; and/or, an area of at least one of the fourth corner sub-protrusion and the fourth side sub-protrusion of the adjacent-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion; wherein an area of the second side sub-protrusion of the separated-type first side protrusion is smaller than an area of the second side sub-protrusion of other first side protrusion; and/or, an area of the fourth side sub-protrusion of the separated-type second side protrusion is smaller than an area of the fourth side sub-protrusion of other second side protrusion; wherein: a height of the second side sub-protrusion of the separated-type first side protrusion in a direction perpendicular to the side where the separated-type first side protrusion is located is smaller than a height of the second side sub-protrusion of other first side protrusion in the direction perpendicular to the side where the other first side protrusion is located; and a height of the fourth side sub-protrusion of the separated-type second side protrusion in a direction perpendicular to the side where the separated-type second side protrusion is located is smaller than a height of the fourth side sub-protrusion of other second side protrusion in the direction perpendicular to the side where the other second side protrusion is located.

19. The touch panel according to claim 18, wherein: a side of the separated-type first side protrusion facing an adjacent first touch electrode has a fifth excision portion; and/or, a side of the separated-type second side protrusion facing an adjacent second touch electrode has a sixth excision portion.

20. The touch panel according to claim 18, wherein: the height of the second side sub-protrusion of the separated-type first side protrusion in the direction perpendicular to the side where the separated-type first side protrusion is located is one-sixth to one-half of the height of the second side sub-protrusion of other first side protrusion in the direction perpendicular to the side where the other first side protrusion is located; and the height of the fourth side sub-protrusion of the

separated-type second side protrusion in the direction perpendicular to the side where the separated-type second side protrusion is located is one-sixth to one-half of the height of the fourth side sub-protrusion of other second side protrusion in the direction perpendicular to the side where the other second side protrusion is located.
